The Beach BIA: Modeling Human Activity and Community Usage

Explanatory Modeling, Clustering, and Bayesian Inference

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derek-slides/beach-uoft.png

Project Objectives

- **Economic Vitality:** Understand foot traffic in the Beaches to increase activity for local businesses.
- **Data:** Leveraging Toronto Open Data to justify BIA budgetary decisions.

Specific Research Questions

1. Can we segment the people coming to the Beaches based off their habits and where they come from?
2. How long do pedestrians stay at the Beaches for?
3. How do external variables (weather, holidays) probabilistically impact pedestrian flow?
4. How these patterns may affect commercial vitality along the strip

Data

Case for Open-Source

The Beach BIA deliberately avoided commercial mobile phone data, relying exclusively on municipal datasets for The Beach BIA.

- **Privacy & Trust:** Eliminates re-identification risks, vulnerable demographics (youth/seniors) inherently under-represented by mobile tracking.
- To avoid expensive purchasing of data, municipal datasets are purpose-built for urban planning and already tax-funded.

Datasets

We aggregated multiple open-source domains from **2021–2025** into a single analytical dataset.

- Extracted Open Data Toronto Bike Share records, strictly filtering for trips originating or terminating at Beach BIA docks.
- Merged the core bike data with Historical Climate Data (Environment Canada), Toronto Beach Water Quality metrics.

Identifying Patterns: Unsupervised Learning

- **Method:** K-Means clustering was used to create "Trip Profiles" by clustering the data based on trip duration and the hour of the day the trip started. Used elbow method to determine optimal number of clusters at 3.

kmeans.png

Cluster Attributes

Table: Cluster Profiles

Cluster	Total Trips	Avg Start Hour	Avg Duration (Mins)	Pct Annual	Pct Casual
1	40243	15.5	61.8	9.9	90.1
2	98882	10.7	20.9	34.0	66.0
3	150026	17.9	21.5	30.5	69.5

Dwell Times

- Want to quantify dwell time in the Beaches.
- **Issue:** There is no way to track individuals, only individual bike rides.

dwell1.png

Dwell Time Analysis

Table: Flow Distribution Statistics

Direction	Average Hour of Day	Median Hour of Day	Total Trips
Inbound	13.99420	14	58575
Outbound	15.41696	16	45937

- Can approximate dwell time by taking difference in average arrival vs departure.
- Average Dwell time is 1.42 hours, approximately 85 minutes.
- Important to note that 21.6% of riders do not bike back, likely leading to longer true dwell time.

Cluster Analysis via Dwell Times

dwell2.png

dwell3.png

Heatmaps

- Want to understand where people are coming from.
- Most trips are coming in from the East (towards downtown).
- Most common destination is Northern Dancer Blvd/Lake Shore, by Woodbine Beach and Park.

heatmaps.png

Bayesian Model Prediction Accuracy

preds.png

Figure: Predictive Accuracy of Bayesian Model

- Model trained on 2021-2024 data, predicting on 2025 data.
- Able to generate accuracte predictions for most days.
- See more inaccuracy on higher counts, likely that these days are holidays and see much different traffic than usual.

Exploratory Data Analysis: Climate and Bike Share Trips

cycle_temp.png

GAMs: Addressing Autocorrelation

Lack of Independence

- Traditional GLMs assume all observations are strictly independent. For daily ridership, treating sequential days as independent ignores inherent time-series momentum, leading to severe residual autocorrelation.
- We correct this by embedding a 1-day lagged response (Trips_{t-1}) directly into the Negative Binomial structure, we explicitly model the temporal dependence:

$$\log(\mu_t) = \text{Environmental \& Calendar Predictors}_t + \phi(\text{Trips}_{t-1})$$

GAMs

Model Diagnostics

- As seen in the Autocorrelation Function (ACF) plots below, the GAM eliminates the severe serial correlation present in the traditional approach.

acf_good_plot.png

acf_bad_plot.png

GAM Results: Drivers of Ridership

$$\log(\mu_t) = \beta_0 + s(\text{Max Temp}_t) + \beta_1(\text{Light}_t) + \beta_2(\text{Precip}_t) + \gamma(\text{Day}_t) + \phi(\text{Trips}_{t-1})$$

Parametric Predictor	**Estimate**	**Std. Error**	**p-value**
Intercept	3.598	0.255	< 0.001
Usable Light (Hours)	0.094	0.018	< 0.001
Precipitation (Hours)	-0.070	0.008	< 0.001
Saturday	0.265	0.091	0.004
Sunday	0.322	0.088	< 0.001
Lagged Trips ($t-1$)	0.0004	0.0001	0.012
Smooth Term	**EDF**	χ^2	**p-value**
Max Temperature (°C)	3.89	540.5	< 0.001

Table: *

Deviance Explained: 87.4% Adj. $R^2 = 0.73$ $n = 302$

Seasonality of the Beach BIA

bike_season1.png

bike_season.png

Seasonal Dates

Isolating Seasonal Dates

- The Negative Binomial GAM strips away daily weather fluctuations to reveal the true underlying ridership trend, incorporating a 'day of year' variable to examine its direct effect.
- By using a restricted smooth spline ($k = 5$), we force a clean curve to mathematically pinpoint the Summer Peak and the steep transitions of the Shoulder seasons.

Phase	Date (2025)	Typical High	Temperature Range
Annual Minimum (Trough)	February 15	-3.6°C	-5.4°C to -2.0°C
Peak Spring Growth	May 2	15.8°C	10.8°C to 19.3°C
Peak Summer Volume	July 15	29.6°C	28.7°C to 32.0°C
Autumn Deceleration	October 14	16.6°C	14.3°C to 18.0°C

(Roughly - On Season: May 2 to October 14, Shoulder Season: February 15 to May 2)

Bayesian GAM: Model Structure & Results

Model Formulation

$$\log(\mu_t) = \beta_0 + s(\text{Max Temp}_t) + \beta_1(\text{Precip}_t) + \gamma(\text{Day Category}_t) + \epsilon_t$$

$$\epsilon_t = \rho\epsilon_{t-1} + \nu_t \quad \text{(AR1 Residual Autocorrelation)}$$

Posterior Estimates (Poisson Family)

Predictor	Estimate	Error	95% Credible Interval
Intercept	4.88	0.04	$[4.81, 4.95]$
Precipitation (Hours)	-0.07	0.01	$[-0.09, -0.06]$
Saturday	0.29	0.07	$[0.16, 0.42]$
Sunday	0.43	0.06	$[0.31, 0.56]$
Statutory Holiday	0.78	0.13	$[0.53, 1.03]$
AR(1) Autocorrelation (ρ)	0.46	0.07	$[0.33, 0.59]$

Table: *

Model converged successfully across 4 chains ($\hat{R} = 1.00$).

Factors for Ridership: Weather & Calendar

temp-bayes.png

Exponential growth as weather warms.

rain_bayes.png

A 5-hour morning rain cuts volume by nearly half.

ride_day_bayes.png

- **Weekday Baseline:** Consistent, lower-volume baseline.
- **Sunday Dominance:** Outperforms Saturdays, making it the highest-priority operational day.

Insights for the Beach BIA

Key Driver	Daily Impact	Business Application
Statutory Holiday	**+118%**	Particularly relevant for them to capture populations who visit the BIA, such as July 1st, for fireworks near Woodbine Beach.
Sunday Traffic	**+53%**	The highest performing day. Target Sunday afternoons for major BIA activations.
Saturday Traffic	**+33%**	A strong baseline weekend increase, but distinctly secondary to Sunday volume.
Hourly Rainfall	**-6.8%**	BIA can tend to expect decreased traffic based on morning weather forecasts.

Bayesian GAM Model: Casual vs. Annual Riders

user_type_temp.png

user_type_rain.png

Calendar Impact: Casual vs. Annual Riders

user_type_day.png